

Wiederholung: Cauchy'sche Prod.formel.

$$\sum_{n=0}^{\infty} a_n \cdot \sum_{n=0}^{\infty} b_n : \text{Konv. (abs.)}$$

$$\Rightarrow \sum_{n=0}^{\infty} c_n, \text{ mit } \sum_{k=0}^n a_k b_{n-k} =: c_n$$

$\Leftrightarrow$ Konv. (abs.)

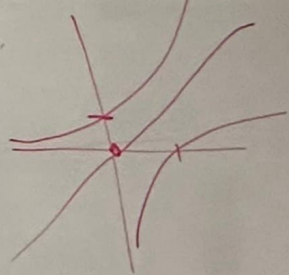
$$\left(\sum_{n=0}^{\infty} a_n \right) \cdot \left(\sum_{n=0}^{\infty} b_n \right) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right)$$

P5.1(a) $\sum_{n=0}^{\infty} q^n, |q| < 1$

$$\sum_{n=0}^{\infty} q^n \cdot \sum_{m=0}^{\infty} q^m = \sum_{n=0}^{\infty} \left(\sum_{k=1}^n q^k \cdot q^{n-k} \right)$$

$$q^{k+n-k} = q^n$$

$$= \sum_{n=0}^{\infty} (n+1) q^n$$



$$(b) \sum_{n=0}^{\infty} n q^n = \sum_{n=0}^{\infty} (n+1) q^n - \sum_{n=0}^{\infty} q^n = \left(\sum_{n=0}^{\infty} q^n \right)^2 - \sum_{n=0}^{\infty} q^n$$

$$|q| < 1$$

geo m.
Summen-
formel.

$$\left(\frac{1}{1-q} \right)^2 - \frac{1}{1-q} = \frac{1-q - (1+q^2 - q)}{(1-q)^3} = \frac{q(q-1)}{(q-1)^3} = \left(\frac{q}{(q-1)^2} \right)$$

$$(c) 0 + \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \dots = \sum_{n=0}^{\infty} \frac{n}{2^n} = \sum_{n=0}^{\infty} n \cdot \left(\frac{1}{2} \right)^n \stackrel{(b)}{=} \frac{1/2}{1/4} = 2$$

Wiederholung II: Konv. radius

Formeln zur Berechnung von K.R.
(Wurzelk-rit.)

Cauchy-Hadamard:

$$\rho = \frac{1}{\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}}$$

Euler/Quotientenk-rit.

$$\rho = \frac{1}{s} \quad s := \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$$

falls s existiert.

T 5.2 (a) $\sum_{n=0}^{\infty} \frac{n^2}{2^n} \cdot \frac{1}{z^n}$

I. Cauchy-Hadamard:

$$\rho = \frac{1}{\limsup_{n \rightarrow \infty} \left(\frac{n^2}{2^n} \right)^{1/n}} = \left(\limsup_{n \rightarrow \infty} \frac{n^{2/n}}{2^{1/n}} \right)^{-1}$$

Beh: $a_n \xrightarrow{n \rightarrow \infty} 1$

$\underline{ZZ}$: Sei $\varepsilon > 0$ bel. $\exists N \in \mathbb{N}$

$\forall n > N: |a_n - 1| < \varepsilon$

Bew: $a_n = n^{2/n} = e^{\frac{2}{n} \ln(n)} = e^{2 \left(\frac{\ln(n)}{n} \right)}$

$$\lim_{n \rightarrow \infty} e^{2 \left(\frac{\ln(n)}{n} \right)} = e^{2 \cdot 0} = 1$$

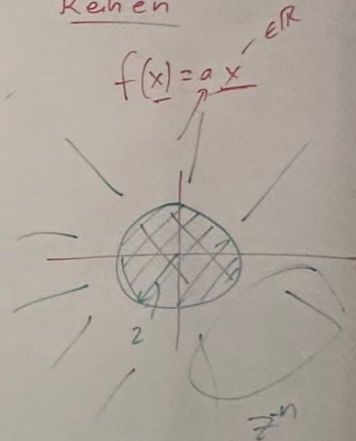
T 5.2 (b) $z + z^2 + z^4 + z^8 + \dots = \sum_{n=0}^{\infty} z^{2^n} = \sum_{n=0}^{\infty} a_n z^n$

$$a_n = \begin{cases} 1, & \exists k \in \mathbb{N}: n = 2^k \\ 0, & \text{sonst} \end{cases}$$

$$\rho = \frac{1}{\limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}} = 1$$

T 5.3 (a)

Laurent Reihen



5.3: (a) $K \in \mathbb{Z}$, $q \neq 0$

$$\sum_{n=0}^{\infty} \frac{n^K}{q^n} \geq n$$

$$\begin{aligned} S &:= \lim_{n \rightarrow \infty} \sup \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} \sup \left(\frac{n^K}{q^n} \right)^{1/n} \\ &= \frac{1}{q} \cdot \lim_{n \rightarrow \infty} n^{K/n} \left(e^{\frac{K \ln(n)}{n}} \right) \\ &= \frac{1}{q} \end{aligned}$$

$$\Leftrightarrow \rho = \frac{1}{S} = q$$

$$\rightarrow q > 1 \Rightarrow \sum_{n=0}^{\infty} \frac{n^K}{q^n} \text{ konvergiert absolut.} \quad q > 1 \rightarrow \frac{n^K}{q^n} \rightarrow 0$$

$$\sum_{n=0}^{\infty} \left(\frac{1}{q} \right)^n = \sum_{n=0}^{\infty} \omega^n, \quad |\omega| = \left| \frac{1}{q} \right| < 1$$

$$\Rightarrow |q| > 1$$

$$(b) \sum_{n=0}^{\infty} \frac{n!}{n^n} \geq n$$

$$\text{Euler: } S = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)!}{n^n} \cdot \frac{n^n}{(n+1)^{n+1}}$$

$$= \lim_{n \rightarrow \infty} \frac{n^n}{(n+1)^n}$$

$$= \lim_{n \rightarrow \infty} \frac{n^n}{n^n} \cdot \frac{1}{\left(1 + \frac{1}{n}\right)^n} = \lim_{n \rightarrow \infty} \left(\left(1 + \frac{1}{n}\right)^n \right)^{-1} = e^{-1}$$

$$\Rightarrow S = \frac{1}{e} \Leftrightarrow \rho = \frac{1}{S} = e$$

Cauchy-Hadamard:

$$\limsup_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{1/n} = \frac{1}{e}$$

$\rightarrow$ für eine positive Folge gilt:

$$\liminf_{n \rightarrow \infty} \frac{1}{a_n} = \frac{1}{\limsup_{n \rightarrow \infty} a_n}$$

$$\hookrightarrow \liminf_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{1/n} = \frac{1}{e}$$

$$\lim_{n \rightarrow \infty} \left(\frac{n!}{n^n} \right)^{1/n} = \frac{1}{e}$$

$$= \lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{n} = \frac{1}{e}$$

$$= \lim_{n \rightarrow \infty} \frac{(n!)^{1/n}}{\left(\frac{n}{e}\right)} = 1$$